RISK 201: General and Other Annuities

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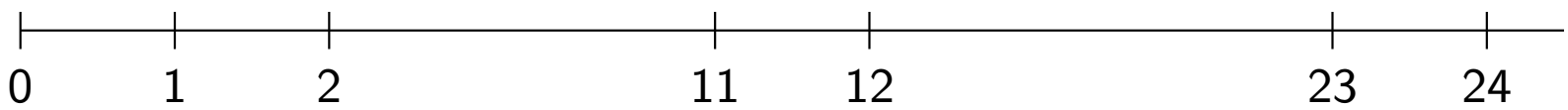
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4.1 General annuities

- An annuity for which payments are made more or less frequently than interest is compounded is called a **general annuity**.
- The simplest technique to solve problems involving general annuities is to **replace the given interest rate** with an equivalent rate whose compounding period matches the payment periods.

Example 4.1

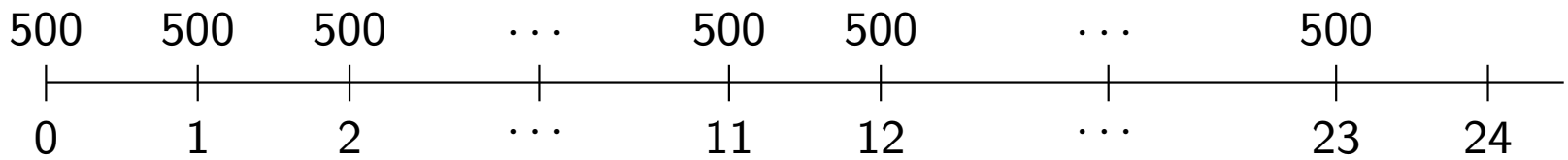
Payments of \$500 are made for two years at the beginning of each month. Find the present value of these payments if the effective annual rate is 6%.



Alternative approach

An alternative approach is to **replace the payments** with equivalent payments whose frequency matches the interest compounding period.

Example 4.1, continued



Ordinary annuity payable m-thly

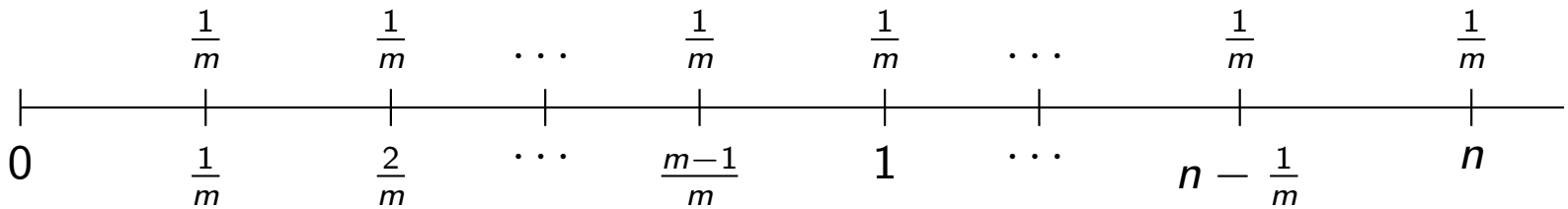
$m = \#$ of payment periods in one interest conversion period,

$n =$ term of the annuity (in interest conversion periods),

$i =$ effective interest rate per period

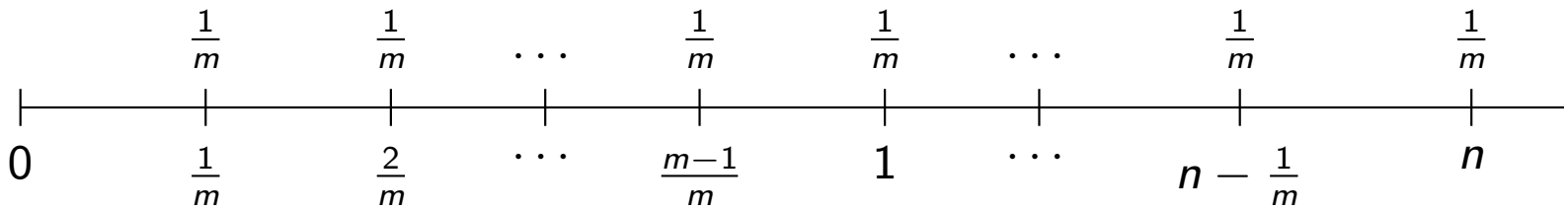
Actuarial notation for present value of an annuity that pays $1/m$ at the end of each m-thly period for n years is $a^{(m)}_{\overline{n}|i}$, read as "a upper m angle n at i".

PV of annuity payable m-thly



$$PV = a^{(m)}_{\overline{n}|i} = \frac{1 - v^n}{i^{(m)}}$$

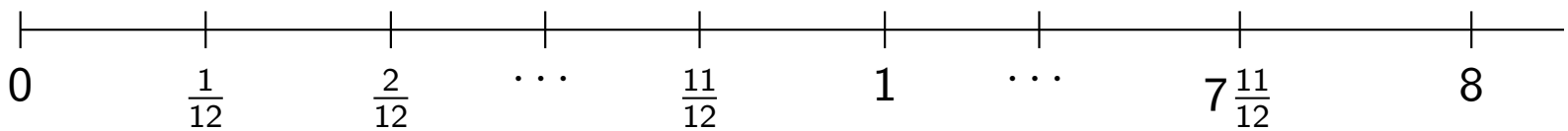
FV of annuity payable m-thly



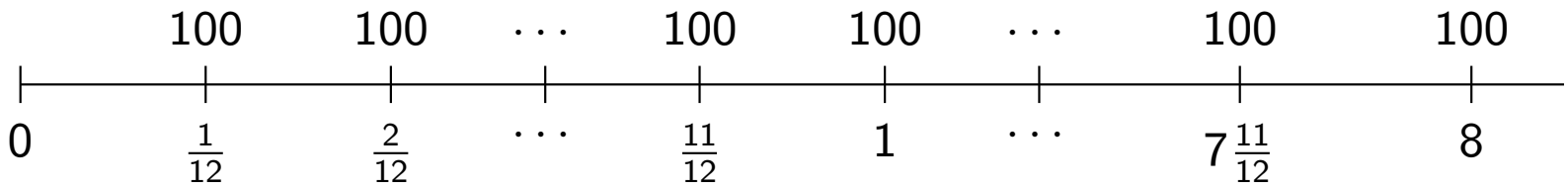
$$FV = s^{(m)}_{\overline{n}|i} = \frac{1 - v^n}{i^{(m)}} \cdot (1 + i)^n = \frac{(1 + i)^n - 1}{i^{(m)}}$$

Example 4.2

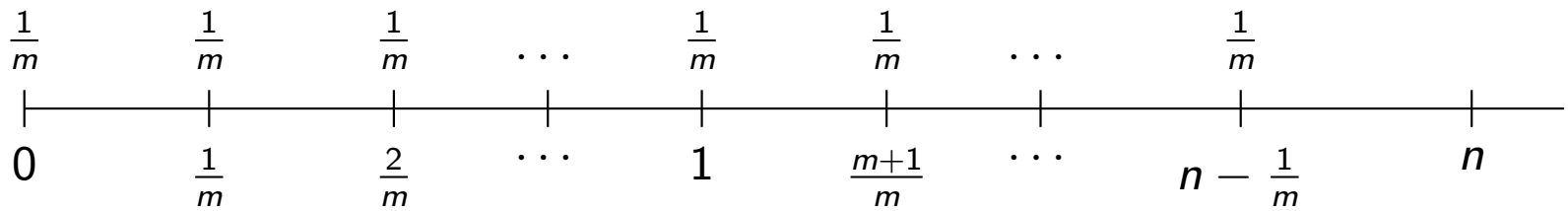
You will receive 96 monthly payments starting one month from now. Payments are \$100 and the effective annual rate of interest is 12%. Find the present value of this annuity today.



Example 4.2, continued



Annuity-due payable m-thly



$\ddot{a}^{(m)}_{\overline{n}|i}$ is PV of annuity that pays $1/m$ at the **beginning** of each m-thly period for n years, $\ddot{s}^{(m)}_{\overline{n}|i}$ is FV

$$\ddot{a}^{(m)}_{\overline{n}|i} = \frac{1-v^n}{d^{(m)}} \qquad \text{and} \qquad \ddot{s}^{(m)}_{\overline{n}|i} = \frac{(1+i)^n - 1}{d^{(m)}}$$

4.2 Mortgages

- Mortgages are a special type of loan where the **property is pledged as security** in case of default (non-payment) by the borrower.
- In Canada, mortgages are offered by various banks, credit unions and private lenders.
- We distinguish between **residential mortgages** (for properties where the owners will live) and **commercial mortgages** (for properties purchased as investment). The processes and rules for issuing commercial mortgages are different from residential mortgages.

Lenders

- **"A" lenders**: Mortgage lenders with strict criteria about who they lend to, heavily regulated
  - The "Big Five" and their various brands
  - Some smaller banks
  - Credit unions
- **"B" lenders**: mortgage lenders with less strict criteria but still regulated
- **"C" lenders**: private lenders, not directly regulated

Regulation

- Banks are regulated federally under the *Bank Act* and supervised by the Office of the Superintendent of Financial Institutions (OSFI)
- Most credit unions are regulated provincially
- *Interest Act* also applies: sets rules about rates and how to communicate/quote them

Down payment

- **Conventional mortgage**: at least 20% down payment (loan-to-value ratio $\leq 80\%$)
- **High-ratio mortgages**: less than 20% down payment (loan-to-value ratio $> 80\%$)
- High-ratio mortgages will typically require mortgage default insurance

Mortgage default insurance

- Mortgage default insurance (aka mortgage loan insurance) protects the lender in case of default by borrower
- Issued by Canada Mortgage and Housing Corporation (CMHC) and a few others
- Only mortgages with a big enough down payment are insured by CMHC

Mortgage default insurance, cont'd

- Minimum down payment to qualify for mortgage insurance through CMHC: 5% of first $500,000 of property value, 10% of rest
- CMHC does not insure mortgages on properties valued over $1.5 / 1 million (homeowner / rental)
- Insurance premium is charged at issue as a % of the mortgage amount, can be paid in cash or rolled into mortgage
- Typical premium is 2.4%–4.0%, depending on LTV

Amortization period

- The period over which the mortgage loan is assumed to be paid back
- Maximum amortization period in Canada:
  - For CMHC-insured mortgages: 30 years
  - For other mortgages: 35 years
- A longer amortization period will:
  - Reduce the amount of mortgage payments
  - Increase the total interest paid over the life of the mortgage

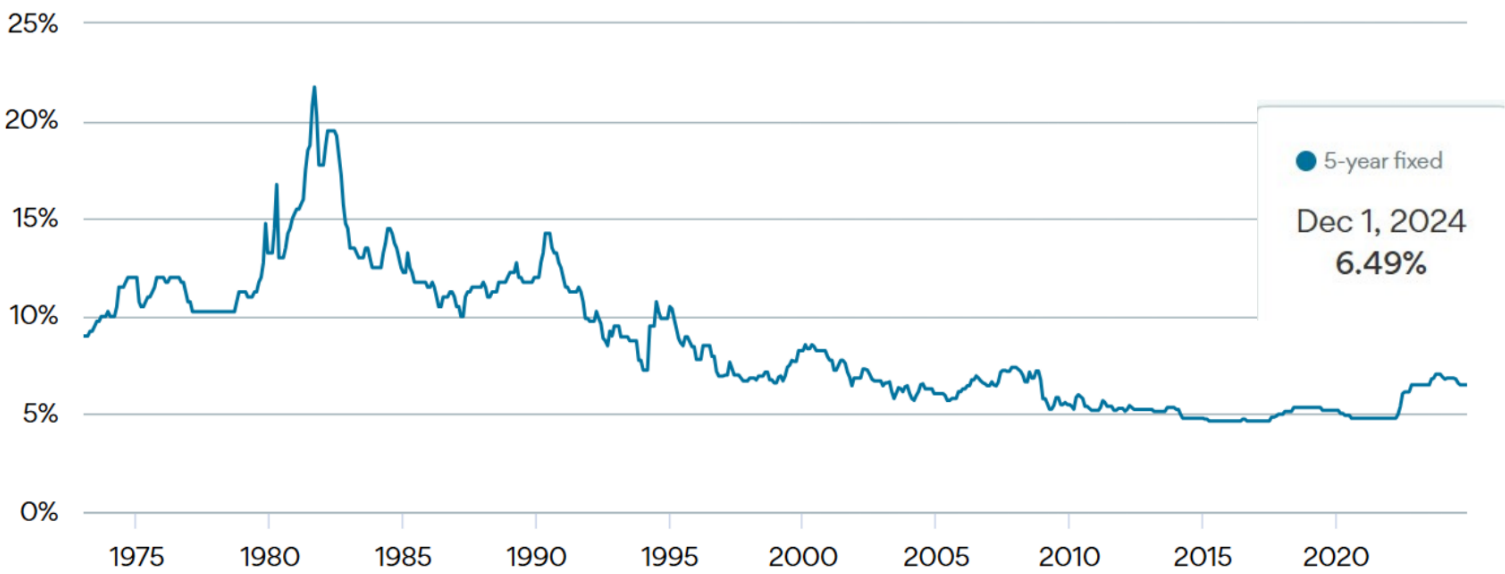
Mortgage rates

- Interest rates payable on mortgages vary widely depending on the type of lender, the credit-worthiness of borrower, and whether the rate is fixed or variable
- Expressed as a nominal rate compounded semi-annually
- Annual Percentage Rate (APR) also includes fees charged to issue the mortgage
- Posted rate vs discounted rate

Fixed vs. variable rates

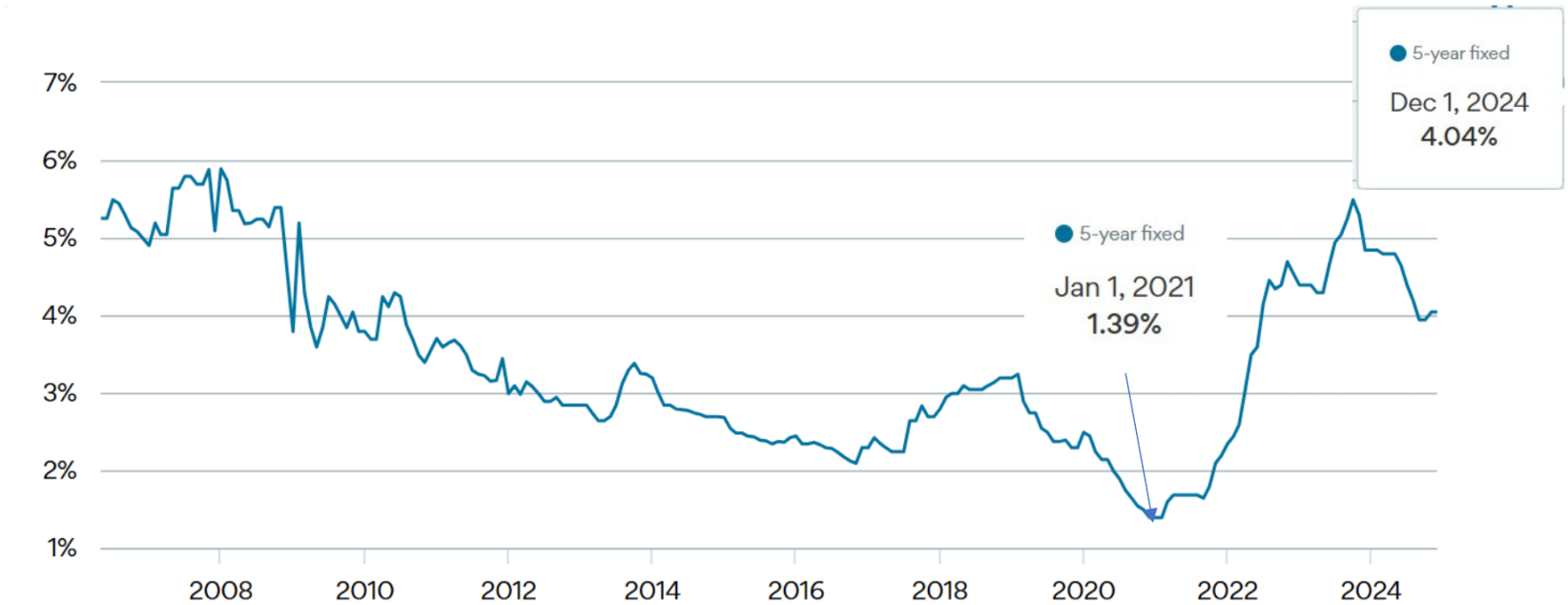
- Fixed mortgage rates are locked in for the length of the mortgage term
- Variable rate floats with the lender's prime rate
- Variable rates on mortgages are generally lower than fixed rates, but borrower takes on more risk

5-year fixed rates – posted



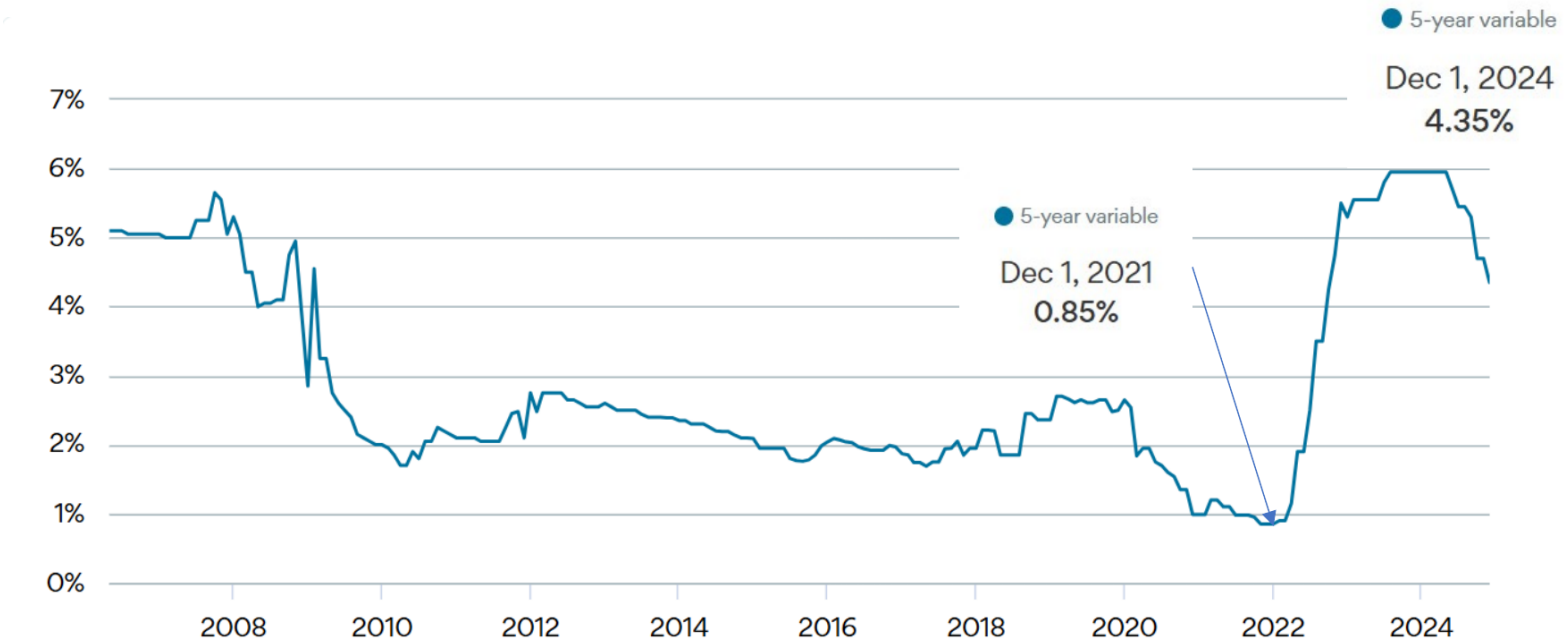
Source: `https://www.ratehub.ca/5-year-fixed-mortgage-rate-history`

5-year fixed rates – discounted



Source: `https://www.ratehub.ca/5-year-fixed-mortgage-rate-history`

5-year variable rates – discounted



Source: `https://www.ratehub.ca/5-year-variable-mortgage-rate-history`

Qualifying for a mortgage

- Credit score
- Loan-to-value ratio
- Income and debt service ratios:
  - Gross debt service ratio (GDS) = PITH/gross income
  - PITH = **P**rincipal, **I**nterest, **T**axes, **H**eat
  - Total debt service ratio (TDS) includes all debt payments in numerator, not just the PITH associated with the mortgage

Mortgage stress testing

- Introduced by OSFI & Dept of Finance in 2012, last updated in June 2021
- Protects stability of financial system in case of rising interest rates
- Calculate GDS and TDS using a **qualifying interest rate**, not the contract rate
- Current qualifying rate is 5.25%
- Maximum GDS $= 39\%$, maximum TDS $= 44\%$

Impact of stress test on borrowing

- As the qualifying rate increases, the monthly mortgage payment that is deemed affordable decreases $\rightarrow$ reduces ability to borrow

- Example 4.3: Consider a borrower with annual household income of \$100,000 and a down payment of \$50,000, wanting a mortgage with a 25-year amortization. Ignore taxes and heating costs for simplicity.

Example 4.3, continued

	Before June 1, 2021	**After June 1, 2021**
Household income	\$100,000	\$100,000
Down payment	\$50,000	\$50,000
Amortization period	25 years	25 years
Qualifying rate	4.79%	5.25%
Maximum monthly pmt	3,250 (=39%*100,000/12)	3,250 (=39%*100,000/12)
Maximum mortgage	$3250\, a^{(12)}_{\overline{25}|0.0479} = 573{,}643$	$3250\, a^{(12)}_{\overline{25}|0.0525} = 548{,}934$
Maximum home value	623,643	598,934

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Prepayments

- The borrower may wish to pay **more** than required under the mortgage terms $\rightarrow$ **prepayment**
- Prepayments accelerate the rate at which principal is repaid and reduce the amount of interest paid during the life of the mortgage
- Lenders put limits on prepayments and charge penalties for prepayments in excess of these limits
  - Fixed rate, closed mortgages have strictest limits
  - Variable rate, open mortgages have no limits

Mortgage-backed securities (MBS)

- The steady stream of payments from mortgage loans makes for a good investment
- Default risk is a potential problem $\rightarrow$ pool many loans together to reduce impact of default by any one borrower
- Lenders sell loans to organizations who will then use these as collateral for a bond issue
- Investors who buy the bond receive monthly payments, secured by the underlying stream of mortgage payments

Types of MBS

- **Pass-through MBS**: payments of interest and principal are passed on to investors directly through a trust
- **Collateralized Debt Obligations (CDOs)**: the loan pool is divided into slices ("tranches") based on characteristics such as time to maturity, credit risk
- Credit rating of CDOs varies widely, depending on characteristics of underlying mortgage pool
- CDOs with bigger risk pay higher rate to investors

CDO-squared

- A **CDO-squared** is a CDO backed by a pool of CDOs
- A way for banks that issue CDOs to spread around their credit risk
- Underlying CDOs divided into tranches (again); priority of different tranches varies in case of default
- Very opaque investment, difficult to assess actual credit risk

"Underwater" mortgages

- When the value of the property is lower than the outstanding balance of the mortgage, the mortgage is said to be underwater
- Can occur in overheated/volatile housing markets following a substantial dip in home values
- Borrowers have incentive to default on obligations and walk away from mortgage
  - "Jingle mail" phenomenon during US subprime mortgage crisis

The US subprime lending crisis

- Low standards for underwriting, increase in housing speculation, mortgage fraud $\rightarrow$ lots of subprime mortgages being issued and packaged into MBS
- Inflated credit ratings assigned to these MBSs by ratings agencies
- When housing bubble burst, borrowers with underwater mortgages defaulted
- Large number of defaults weakened the MBSs written on these pools

Subprime lending crisis, cont'd

- MBS values declined $\rightarrow$ banks/investors losing \$
- Banks unable to issue more loans
  - Credit spreads widened
  - Lending ground to a halt
- Defaults pushed housing prices further down, throwing more fuel on the fire
- Several US banks teetering on brink of insolvency, eventually bailed out by US government and US Federal Reserve Bank

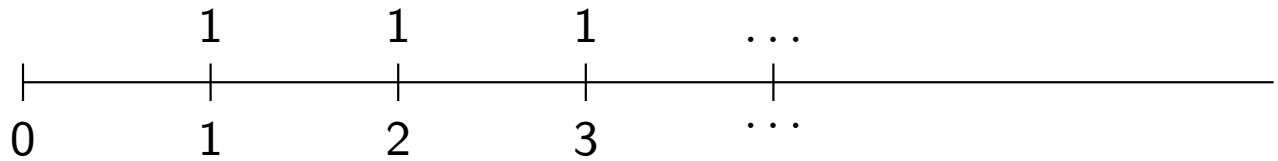
Global financial crisis (2007–2008)

- Losses on MBSs and CDOs and CDO^2 were spread around globally
- Many banks both issuers of CDOs and holders of other banks' CDOs
- Trouble at a small number of banks in the US led to contagion throughout global banking system
- Highlighted interconnectedness of global financial system and led to regulatory focus on Systemically Important Financial Institutions (SIFI)

4.3 Perpetuities

A perpetuity is an annuity whose payments go on **forever**.

Ordinary simple perpetuity (perpetuity-immediate):



$$PV = a_{\overline{\infty}|i} = 1/i$$

PV of ordinary perpetuity

Recall the sum of an infinite geometric series:

$$S = a + ar + ar^2 + ar^3 + \cdots = \frac{a}{1-r}$$

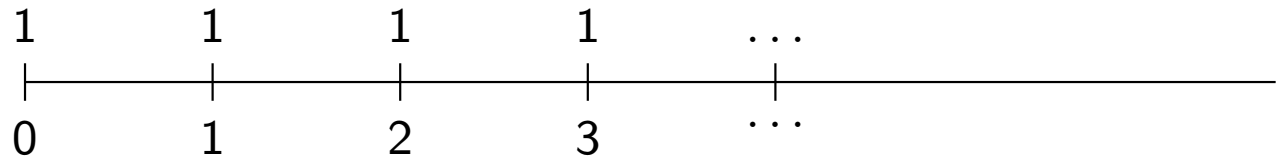
PV of ordinary perpetuity:

$$a_{\overline{\infty}|i} = 1/i$$

Limiting case of an n-year annuity as $n \to \infty$:

$$a_{\overline{\infty}|i} = \lim_{n\to\infty} a_{\overline{n}|i} = \lim_{n\to\infty} \frac{1-v^n}{i}$$

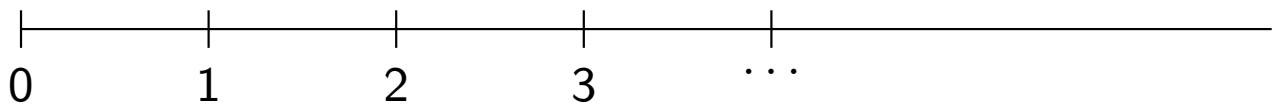
Perpetuity due



$$PV = \ddot{a}_{\overline{\infty}|i} = 1 + v + v^2 + \cdots = 1 + \frac{1}{i} = 1/d$$

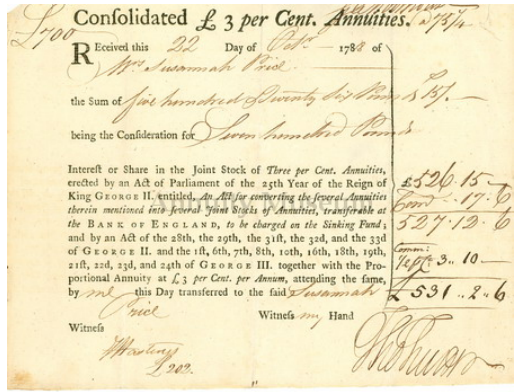
Example 4.4

You want to establish an annual \$2000 scholarship for outstanding actuarial students. If the first scholarship payment is one year from now, how much should you set aside in an endowment fund earning 4% effective?



Practical examples of perpetuities

- Endowment funds
  - Scholarships and bursaries
  - Institutional funding
  - Sovereign wealth funds
- 'Consol' bonds (UK, US)
  - Short for 'consolidated annuities'
  - Borrowing by government, promise to pay fixed % of initial principal each year (coupon) to investor, to be redeemed when government chooses
  - UK consol first issued in 1752, final redemption in 2015
- Preferred shares
- Rental income


Consolidated £ 3 per Cent. Annuities.

Received this 22 Day of Oct. 178? of

Mrs. Susannah Price

the Sum of five hundred Seventy two Pounds £572.—

being the Consideration for Seven hundred Pounds

Intereſt or Share in the Joint Stock of *Three per Cent. Annuities*, erected by an Act of Parliament of the 25th Year of the Reign of King GEORGE II. entitled, *An Act for converting the ſeveral Annuities therein mentioned into federal Joint Stocks of Annuities, tranſferable at the* BANK OF ENGLAND, *to be charged on the Sinking Fund*; and by an Act of the 28th, the 29th, the 31ſt, the 32d, and the 33d of GEORGE II. and the 1ſt, 6th, 7th, 8th, 10th, 16th, 18th, 19th, 21ſt, 22d, 23d, and 24th of GEORGE III. together with the Proportional Annuity at £3 *per Cent. per Annum*, attending the ſame, by Mr. Price this Day transferred to the ſaid Susannah

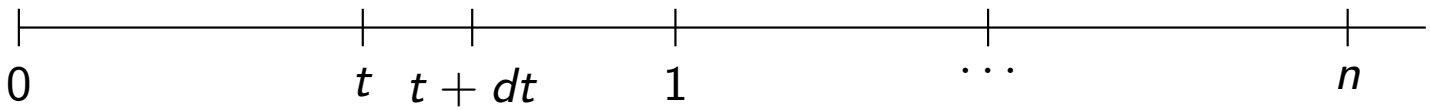
£526 15
Com. 17 6
527 12 6
Comm. 3 10
£531 2 6

Witneſs Witneſs my Hand

Wharton

4.4 Continuous Annuities

The **present value** of an annuity making payments **continuously** at a rate of 1 per interest period for n periods is denoted by $\bar{a}_{\overline{n}|}$.



$$\bar{a}_{\overline{n}|} = \int_0^n v^t \, dt = \int_0^n e^{-\delta t} \, dt$$

Limit of m-thly ordinary annuity

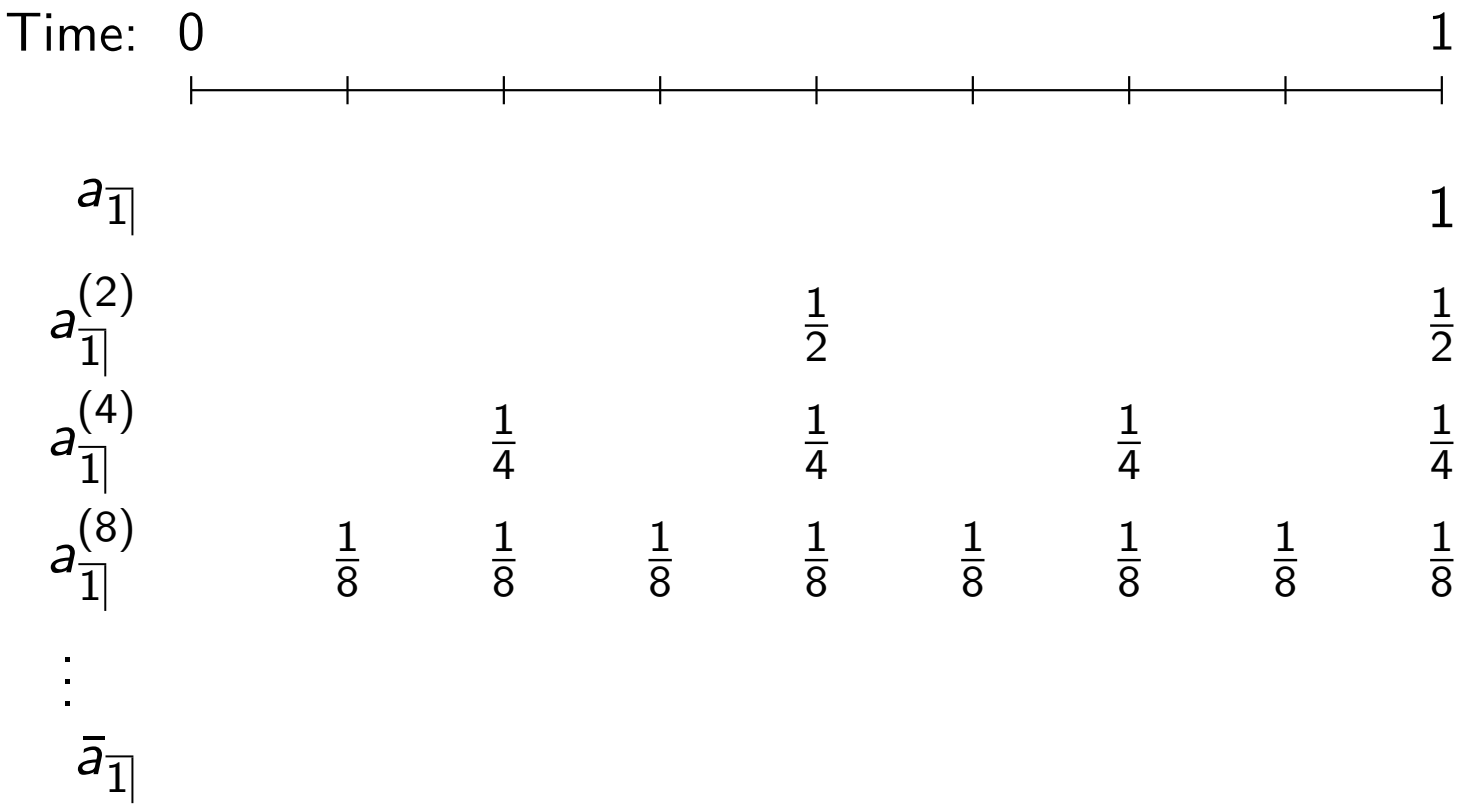
Recall that

$$\lim_{m\to\infty} i^{(m)} = \delta.$$

Then

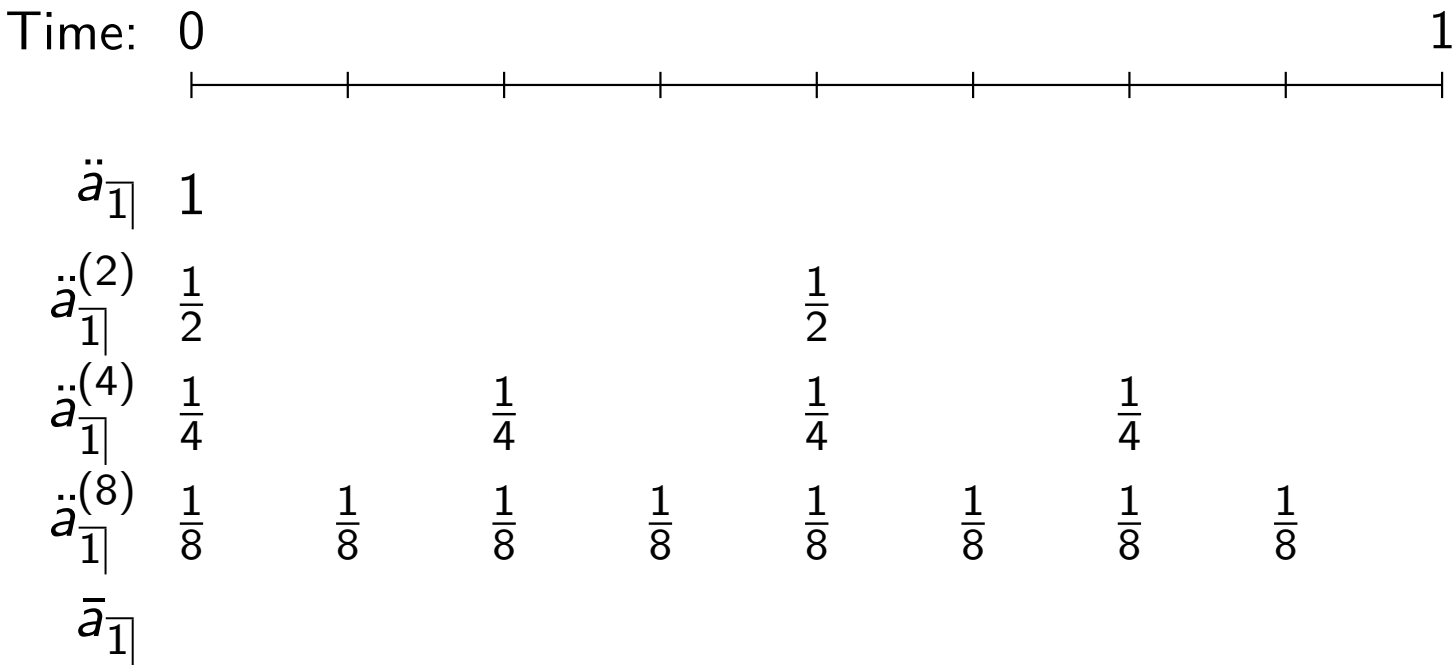
$$\bar{a}_{\overline{n}|} = \frac{1-v^n}{\delta} = \lim_{m\to\infty} \frac{1-v^n}{i^{(m)}} = \lim_{m\to\infty} a_{\overline{n}|}^{(m)}$$

Interpretation ($n = 1$)



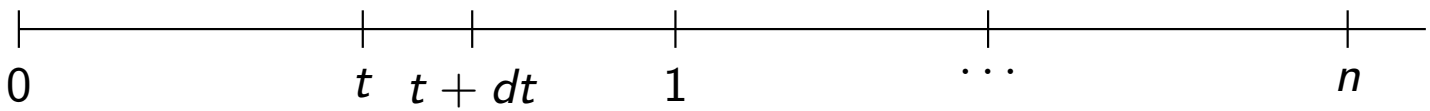
Limit of m-thly annuity-due

$$\lim_{m\to\infty} d^{(m)} = \delta \quad \Rightarrow \quad \bar{a}_{\overline{n}|} = \lim_{m\to\infty} \ddot{a}^{(m)}_{\overline{n}|}$$



Accumulated value

The **accumulated value** of an annuity making payments **continuously** at a rate of 1 per interest period for n periods is denoted by $\bar{s}_{\overline{n}|}$.



$$\bar{s}_{\overline{n}|} = \int_0^n e^{\delta(n-t)}\,dt = e^{\delta n}\int_0^n e^{-\delta t}\,dt = e^{\delta n}\,\bar{a}_{\overline{n}|} = \frac{(1+i)^n - 1}{\delta}$$

Example 4.5

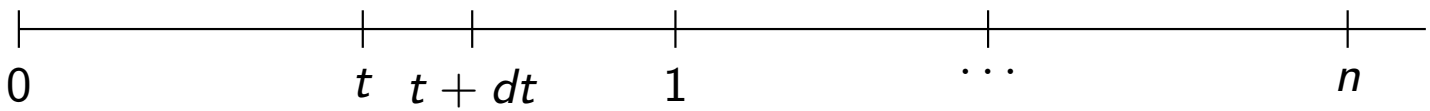
- In 2021 and 2022 Daisy deposits \$12/day into an account.
- The account earns interest from the exact time of the deposit, with interest quoted as an annual effective rate.
- The rates are 9% in 2021 and 2022.
- Find the amount in the account, including interest, on December 31, 2022 in two ways:
  (a) exactly based on the daily deposits, and
  (b) using the approximation that deposits are made continuously.

Example 4.5, continued

Please see the Excel spreadsheet 'daily, monthly, continuous annuities' file on Canvas.

I also recommend you try quarterly / bi-annually / usual annuities and compare their future values with given ones.

Continuous annuity with varying δ



$$\bar{a}_{\overline{n}|} = \int_0^n e^{-\int_0^t \delta_s \, ds} \, dt$$

$$= \int_0^n e^{-\int_0^t \delta \, ds} \, dt = \int_0^n e^{-t\delta} \, dt = \frac{1 - v^n}{\delta} \quad \text{(if } \delta \text{ is constant)}$$

$$\bar{s}_{\overline{n}|} = \int_0^n e^{\int_t^n \delta_s \, ds} \, dt$$

Example 4.6 (Brown & Kopp 4.4.A7)

A level annuity pays \$100 per annum continuously for 20 years. If $\delta_t = \frac{1}{1+t}$, calculate the present value of this annuity.

$$
\begin{aligned}
PV = 100\,\bar{a}_{\overline{20}|} &= 100 \int_0^{20} e^{-\int_0^t \delta_s\,ds}\,dt = 100 \int_0^{20} e^{-\int_0^t \frac{1}{1+s}\,ds}\,dt \\
&= 100 \int_0^{20} e^{-\ln(t+1)}\,dt \\
&= 100 \int_0^{20} 1/(t+1)\,dt = 100 \ln 21
\end{aligned}
$$

4.5 Varying annuities

In many practical situations, we have a payment stream that **varies deterministically** over time.

We will consider three "patterns":

- Discrete payments that vary by a constant %
  - $\rightarrow$ payments form a geometric progression
- Discrete payments that vary by a constant \$
  - $\rightarrow$ payments form an arithmetic progression
- Continuous payments that vary with time

Geometric progressions

In many situations, payments are adjusted by a fixed percentage:

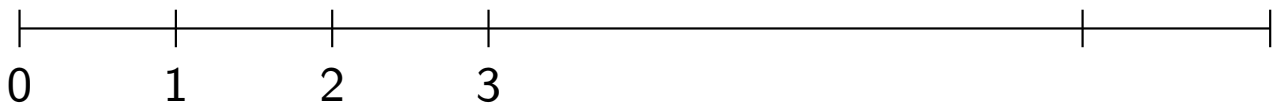
- pensions adjusted for inflation,
- wages increased according to a negotiated agreement.

The adjustments can happen:

- at the same frequency as the payments, or
- at a different frequency.

Example 4.7

A pension plan will make annual payments starting one year from now and continuing for a total of 20 payments. The first payment will be \$24,000, with cost-of-living adjustments of 5% per year applied to each payment thereafter. Find the present value of this pension arrangement today at an effective rate of 10%.



Example 4.7, continued

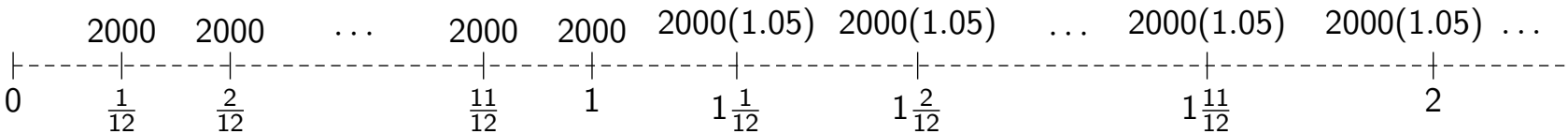
$$\begin{aligned}\text{PV} = v + (1+r)v^2 + (1+r)^2 v^3 + \cdots \\ + (1+r)^{n-2} v^{n-1} + (1+r)^{n-1} v^n\end{aligned}$$

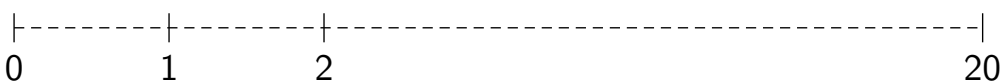
Where $v = \dfrac{1}{1+i}$.

Q: What happens if $i = r$? PV $= nv$

Example 4.8

Suppose the pension plan makes monthly payments of \$2,000 starting one month from now, up to 20 years. The payments still only increase once a year, so that the first 12 are \$2,000, the next 12 are \$2,000(1.05), and so on. Find the present value of this pension arrangement with $i^{(12)} = 5.4\%$.





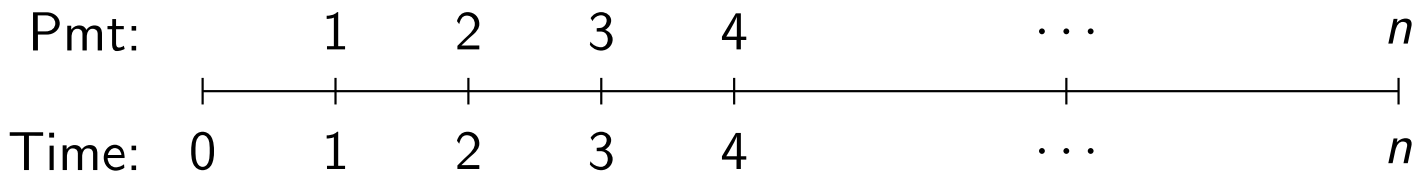
Arithmetic progressions

- Payments are adjusted by a fixed **amount**.
- Formulas can be derived for annuities whose payments vary in arithmetic progression:
  - n-year increasing ordinary annuity

$$PV = (Ia)_{\overline{n}|i}, \quad FV = (Is)_{\overline{n}|i}$$

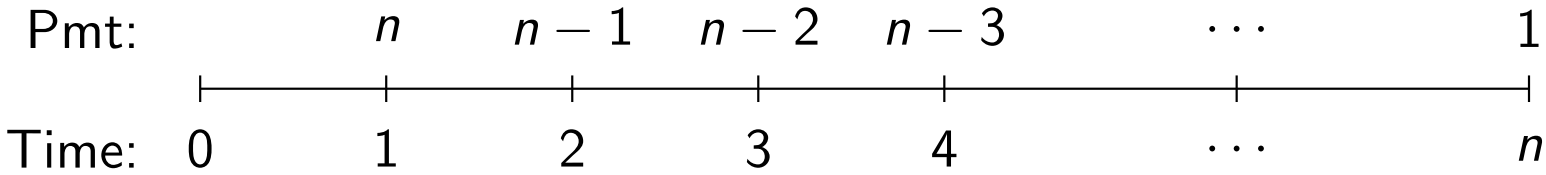
  - n-year decreasing ordinary annuity
  - n-year increasing annuity-due
  - n-year decreasing annuity-due

n-year increasing ordinary annuity



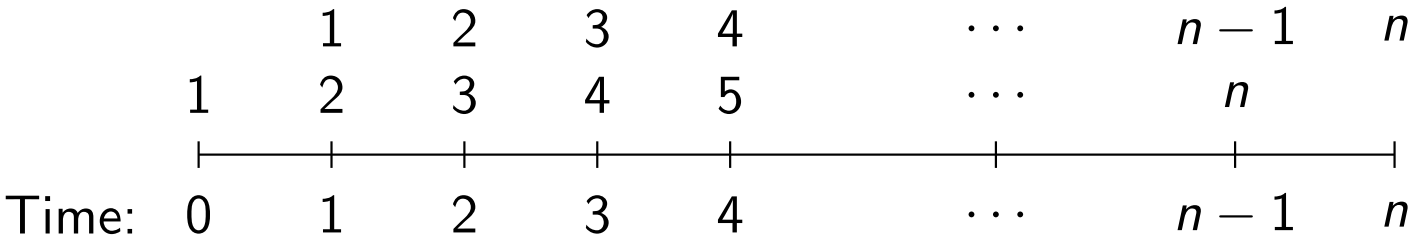
$$
\begin{aligned}
(Ia)_{\overline{n}|i} &= 1v + 2v^2 + 3v^3 + 4v^4 + \cdots + nv^n \\
v\,(Ia)_{\overline{n}|i} &= \qquad 1v^2 + 2v^3 + 3v^4 + \cdots + (n-1)v^n + nv^{n+1} \\
(1-v)(Ia)_{\overline{n}|i} &= v + v^2 + v^3 + \cdots + v^n - nv^{n+1} = a_{\overline{n}|i} - nv^{n+1} \\
\therefore\ (Ia)_{\overline{n}|i} &= \frac{a_{\overline{n}|i} - nv^{n+1}}{1-v} = \frac{(a_{\overline{n}|i} - nv^{n+1})(1+i)}{(1-v)(1+i)} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{i} \\
(Is)_{\overline{n}|i} &= (Ia)_{\overline{n}|i}\,(1+i)^n = \frac{\ddot{s}_{\overline{n}|i} - n}{i}
\end{aligned}
$$

n-year decreasing ordinary annuity



$$
\begin{aligned}
(Da)_{\overline{n}|i} &= nv + (n-1)v^2 + (n-2)v^3 + \cdots + 1v^n \\
\frac{1}{v}(Da)_{\overline{n}|i} &= n + (n-1)v + (n-2)v^2 + \cdots + 1v^{n-1} \\
\frac{1-v}{v}(Da)_{\overline{n}|i} &= n - \left(v + v^2 + \cdots + v^{n-1} + v^n\right) = n - a_{\overline{n}|i} \\
\therefore\ (Da)_{\overline{n}|i} &= \frac{v}{1-v}\left(n - a_{\overline{n}|i}\right) = \frac{n - a_{\overline{n}|i}}{i} \\
(Ds)_{\overline{n}|i} &= (Da)_{\overline{n}|i}(1+i)^n = \frac{n(1+i)^n - s_{\overline{n}|i}}{i}
\end{aligned}
$$

Increasing/decreasing annuity-due



$$(I\ddot{a})_{\overline{n}|i} = (Ia)_{\overline{n}|i}\,(1+i) = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{d}$$

$$(I\ddot{s})_{\overline{n}|i} = (Is)_{\overline{n}|i}\,(1+i) = \frac{\ddot{s}_{\overline{n}|i} - n}{d}$$

$$(D\ddot{a})_{\overline{n}|i} = (Da)_{\overline{n}|i}\,(1+i) = \frac{n - a_{\overline{n}|i}}{d}$$

$$(D\ddot{s})_{\overline{n}|i} = (Ds)_{\overline{n}|i}\,(1+i) = \frac{n(1+i)^n - s_{\overline{n}|i}}{d}$$

Example 4.9

You bought a perpetuity-due with annual payments starting at \$50 and increasing by \$50 each year until the payment amount reaches \$1000. The payments stay at \$1000 thereafter. The effective annual interest rate is 7.5%. Find the PV of this perpetuity.



Continuously varying annuities

The present value of an n-year annuity for which payments are made continuously at a rate of t at time t is denoted by $(\bar{I}\bar{a})_{\overline{n}|i}$. The accumulated value at the end of the n-year period is denoted by $(\bar{I}\bar{s})_{\overline{n}|i}$.

$$(\bar{I}\bar{a})_{\overline{n}|i} = \frac{\bar{a}_{\overline{n}|i} - nv^n}{\delta}$$

$$(\bar{I}\bar{s})_{\overline{n}|i} = \frac{\bar{s}_{\overline{n}|i} - n}{\delta}$$

General continuous annuities

Suppose payments are made continuously for n years at rate $f(t)$ at time t, with force of interest δ_t.

$$\begin{aligned}
\text{PV} &= \int_0^n f(t) \cdot e^{-\int_0^t \delta_s \, ds} \, dt \\
&= \int_0^n f(t) \cdot e^{-\delta t} \, dt \quad \text{(if } \delta \text{ is constant)} \\
\text{AV} &= \int_0^n f(t) \cdot e^{\int_t^n \delta_s \, ds} \, dt \\
&= \int_0^n f(t) \cdot e^{\delta(n-t)} \, dt \quad \text{(if } \delta \text{ is constant)}
\end{aligned}$$

Example 4.10

Deposits are being made into a fund continuously at an annual rate of $200e^t$ at time t.

a) Find the present value of 5 years of payments if interest is credited at 6% per year compounded continuously.

$f(t) = 200e^t,\ \delta = 6\%$

Example 4.10, continued

b) What is the accumulated value of the fund at the end of the third year?